

COMP30850

Dynamic Networks

Derek Greene

UCD School of Computer Science
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Introduction

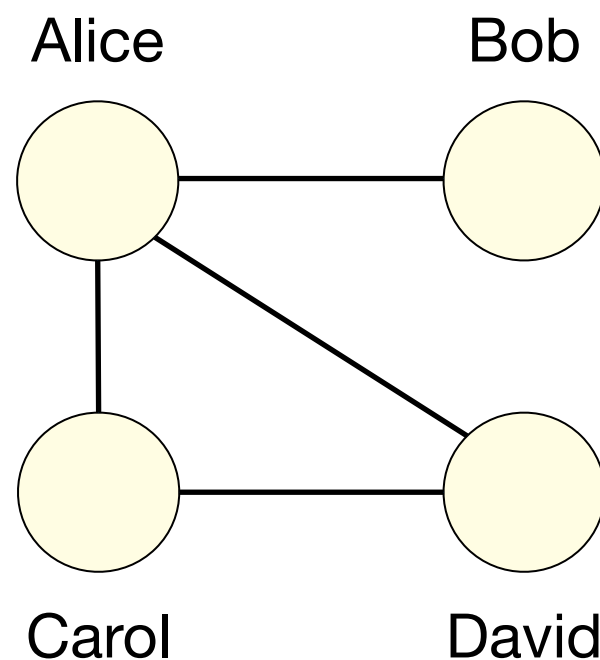
- Many sources of information produce large volumes of **continuous data** which arrives on a streaming basis.
- In such cases, each new piece of data typically has a **timestamp** associated with it, indicating when the data was generated.

Source	Continuous Data
Sensor platform	Readings from one or more sensors
Mobile subscriber network	Call or message logs
Payment systems	Financial transactions and fund transfers
Social media platform	User-generated posts
Code repositories	Code commits to projects

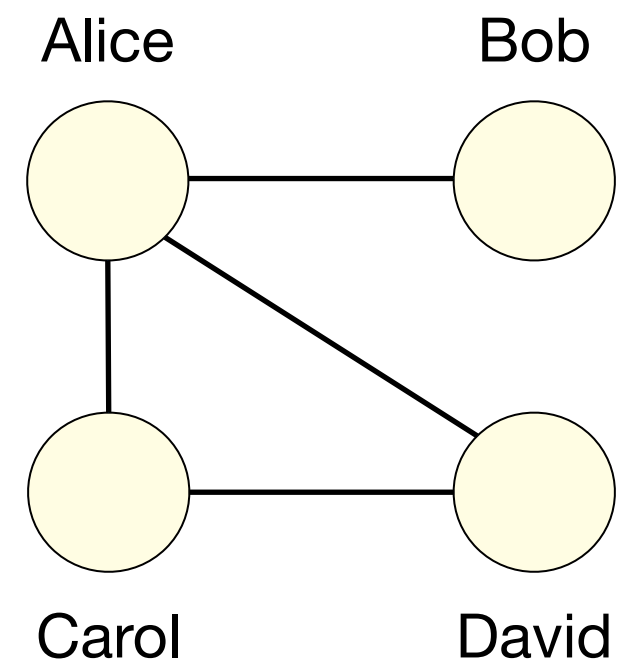
Static Networks

- We often treat network data in a **static** way, ignoring temporal information - i.e., when nodes or edges appear in a network.
- In reality, many networks will change dynamically over time
 - e.g. mobile call networks, Facebook friendships etc.
- Over time the structure of these networks will evolve:
 - Nodes will join and leave the network.
 - Edges between nodes are formed and removed.

A static network representation



Actual evolution of the same network



Dynamic Networks

- **Dynamic network analysis** is the study of change occurring in networks with the passage of time.
- **Example:** Network of mutual friendships between six pupils at the beginning and end of the school year.

	Alison	David	Eliot	Keith	Ross	Sarah
Alison	-	1	0	0	1	0
David	1	-	1	0	0	1
Eliot	0	1	-	0	0	0
Keith	0	0	0	-	1	0
Ross	1	0	0	1	-	1
Sarah	0	1	0	0	1	-

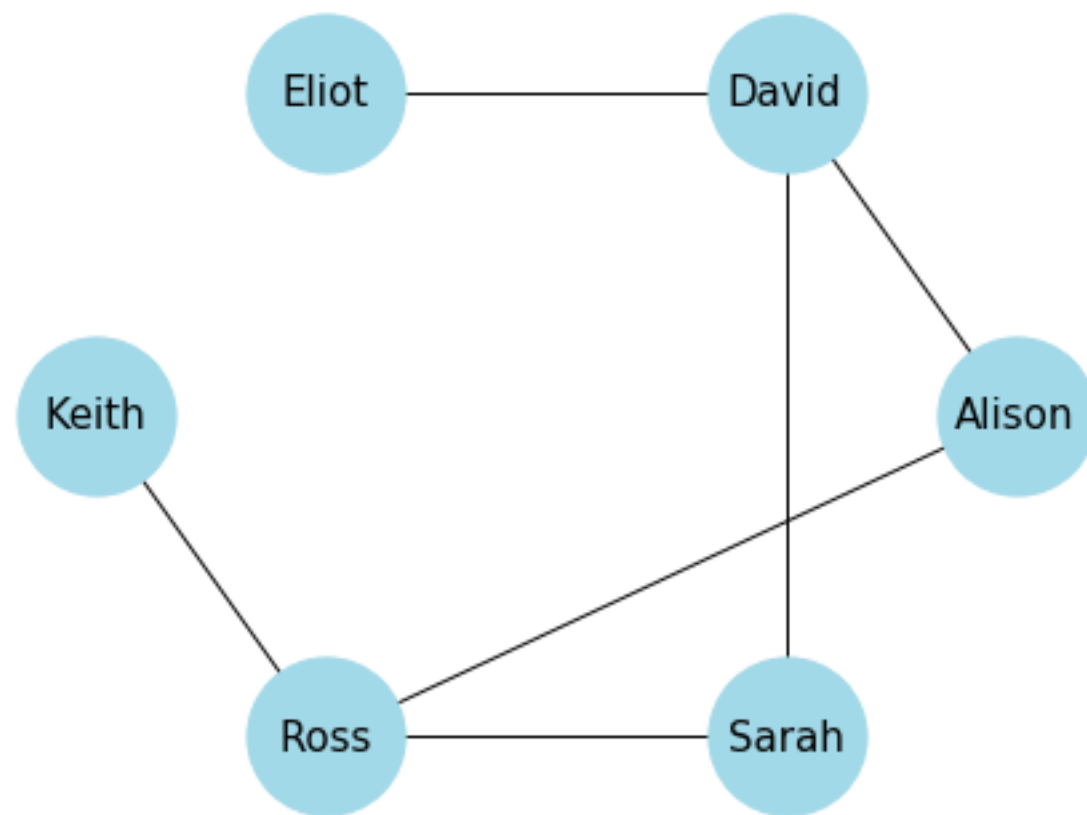
Beginning of school year
6x6 adjacency matrix

	Alison	David	Eliot	Keith	Ross	Sarah
Alison	-	1	0	0	1	0
David	1	-	1	1	1	1
Eliot	0	1	-	0	1	0
Keith	0	1	0	-	1	0
Ross	1	1	1	1	-	1
Sarah	0	1	0	0	1	-

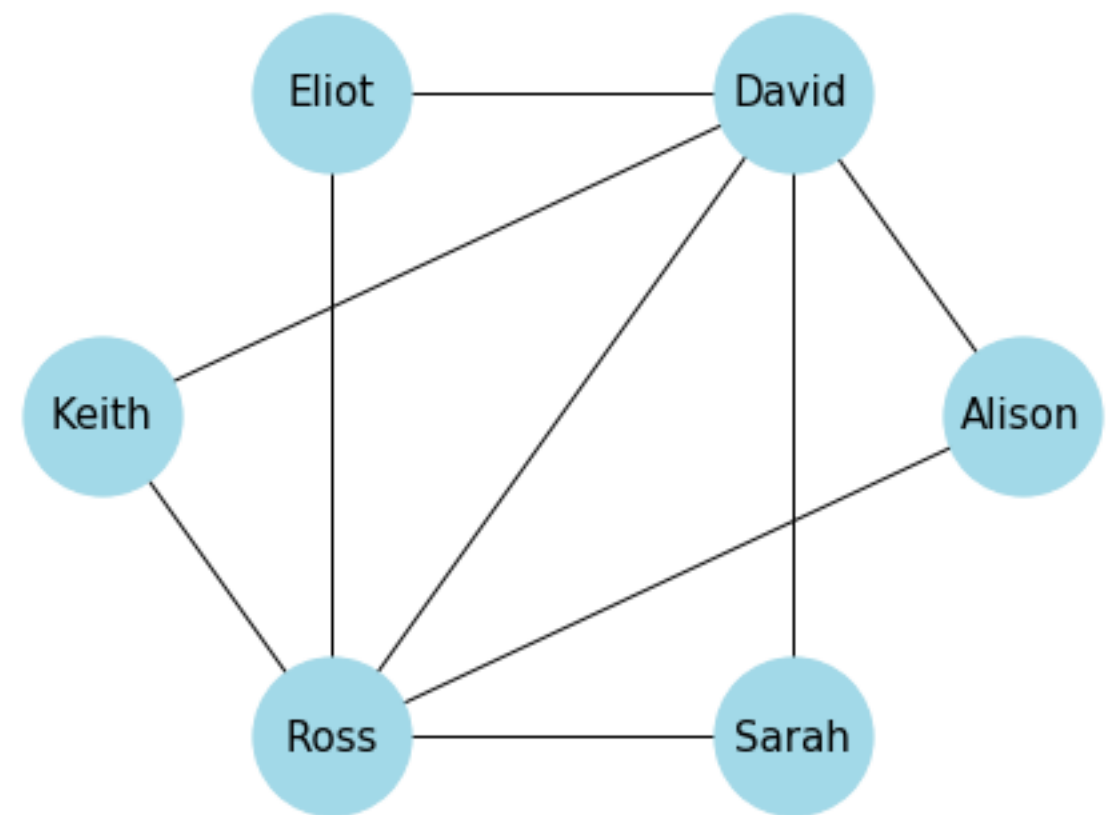
End of school year
6x6 adjacency matrix

Dynamic Networks

- **Dynamic network analysis** is the study of change occurring in networks with the passage of time.
- **Example:** Network of mutual friendships between six pupils at the beginning and end of the school year.



Beginning of school year
Undirected network
6 nodes, 6 edges



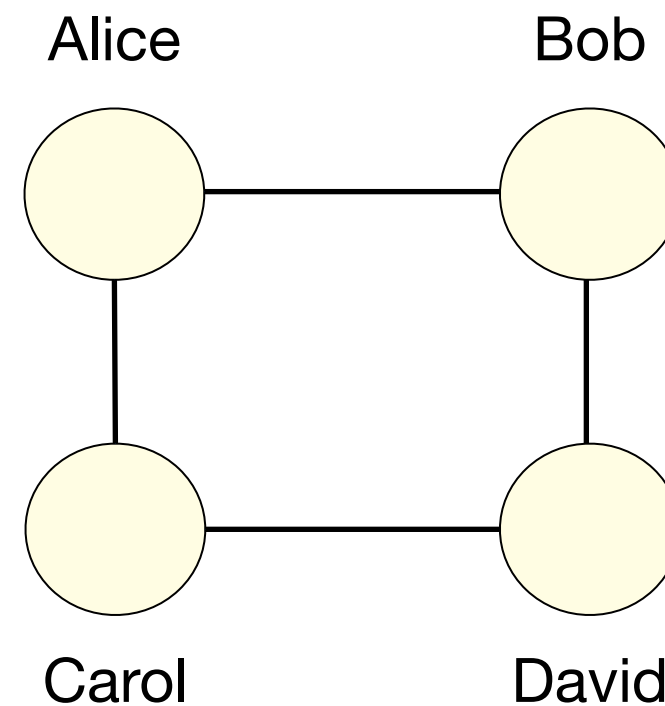
End of school year
Undirected network
6 nodes, 12 edges

Dynamic Representations

- Dynamic network data is often stored as an edge list, with each edge linked to a **timestamp**. This timestamp may represent a specific date and time or an abstract time step.
- Static network analysis methods typically ignore these timestamps.

Day	Node 1	Node 2
Monday	Alice	Bob
Wednesday	Alice	Carol
Thursday	Alice	Carol
Friday	Bob	David
Saturday	Carol	David

Set of five interactions across six days, involving four individuals



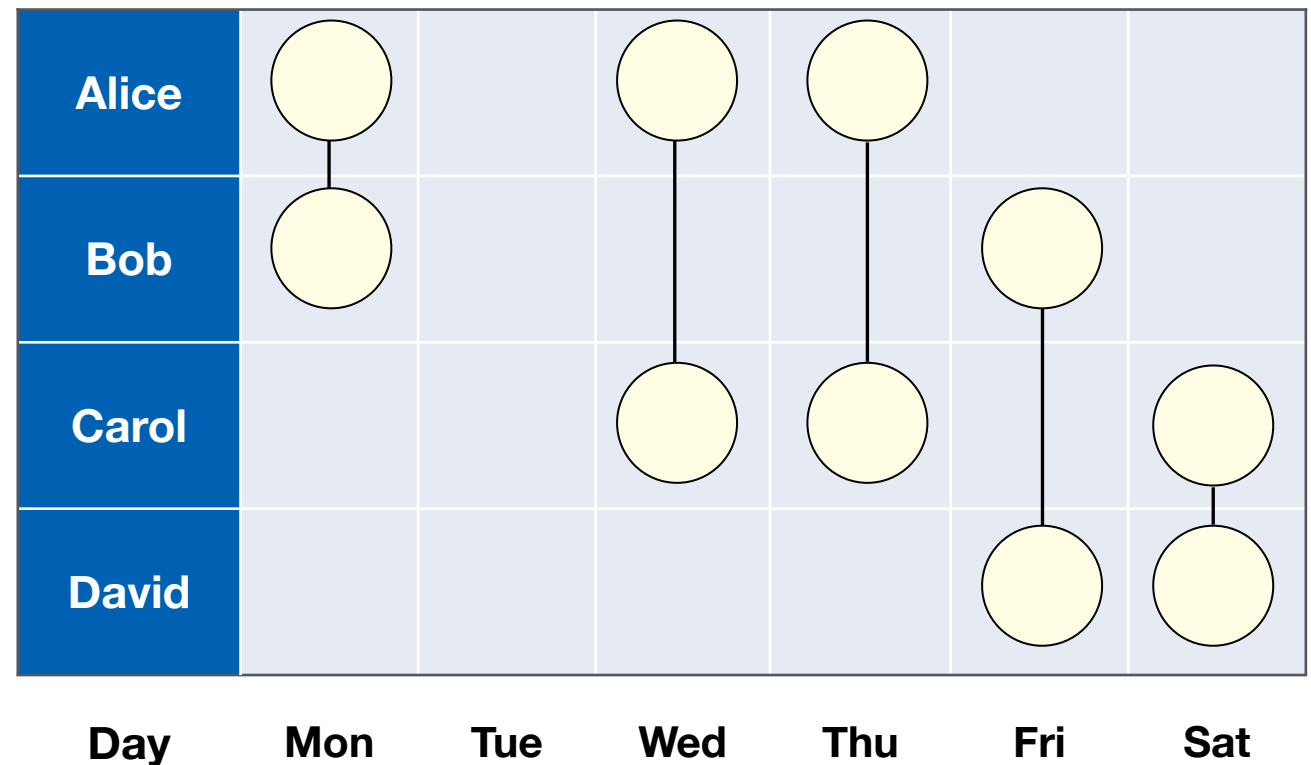
Static network representation of interactions

Dynamic Representations

- Dynamic network data is often stored as an edge list, with each edge linked to a **timestamp**. This timestamp may represent a specific date and time or an abstract time step.
- Static network analysis methods typically ignore these timestamps.

Day	Node 1	Node 2
Monday	Alice	Bob
Wednesday	Alice	Carol
Thursday	Alice	Carol
Friday	Bob	David
Saturday	Carol	David

Set of five interactions across six days, involving four individuals



Timeline representation of individual interactions by timestamp (day)

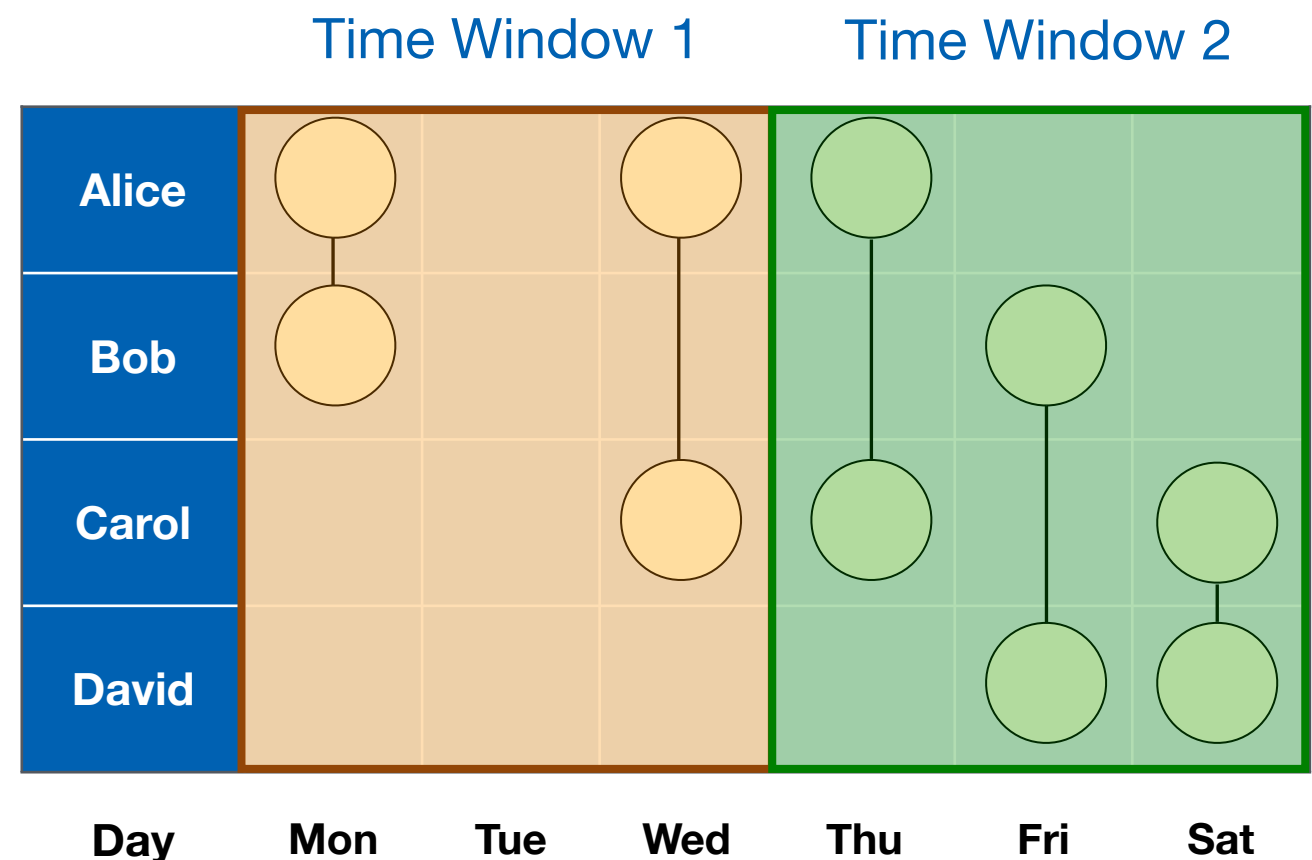
Time Windows

- We could treat each edge as it arrives and build the network incrementally, but this approach poses scalability challenges and makes comparisons over time difficult.
- A more practical approach is to divide continuous data into consecutive, fixed-size **time windows** of equal duration, allowing us to study how the network evolves over distinct periods.

Divide the full set of interactions into two time windows, each covering three timestamps (days)

Window 1: Monday to Wednesday

Window 2: Thursday to Saturday



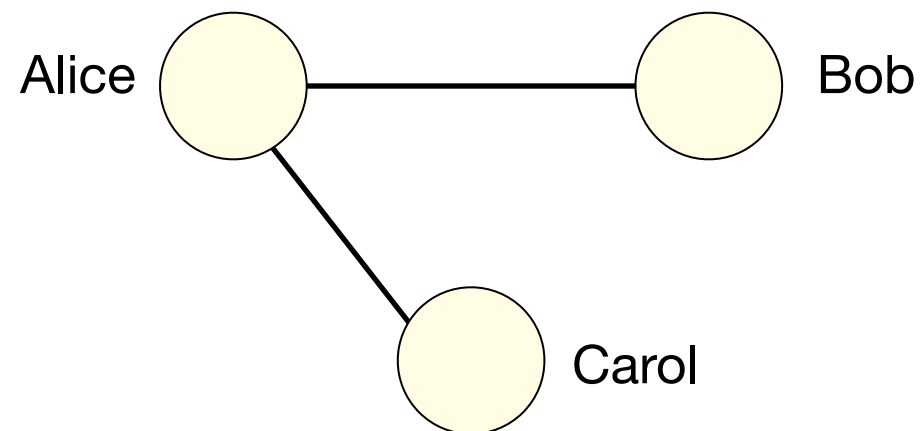
Time Window Networks

- Once the data has been divided into time windows, we create a **time window network** for each window. Each network provides an aggregated snapshot showing all nodes and edges that were active during that period.

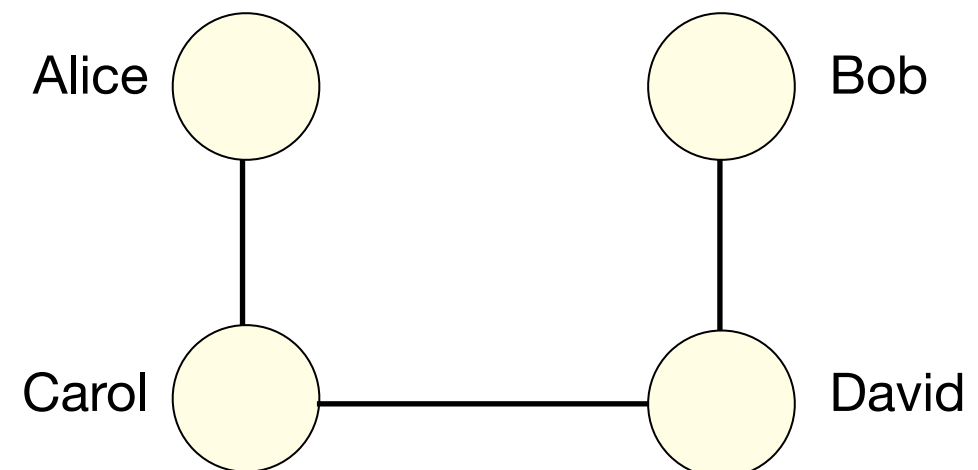
Day	Node 1	Node 2
Monday	Alice	Bob
Wednesday	Alice	Carol
Thursday	Alice	Carol
Friday	Bob	David
Saturday	Carol	David

Divide raw data into two windows:
Window 1: Monday to Wednesday
Window 2: Thursday to Saturday

Time Window Network 1

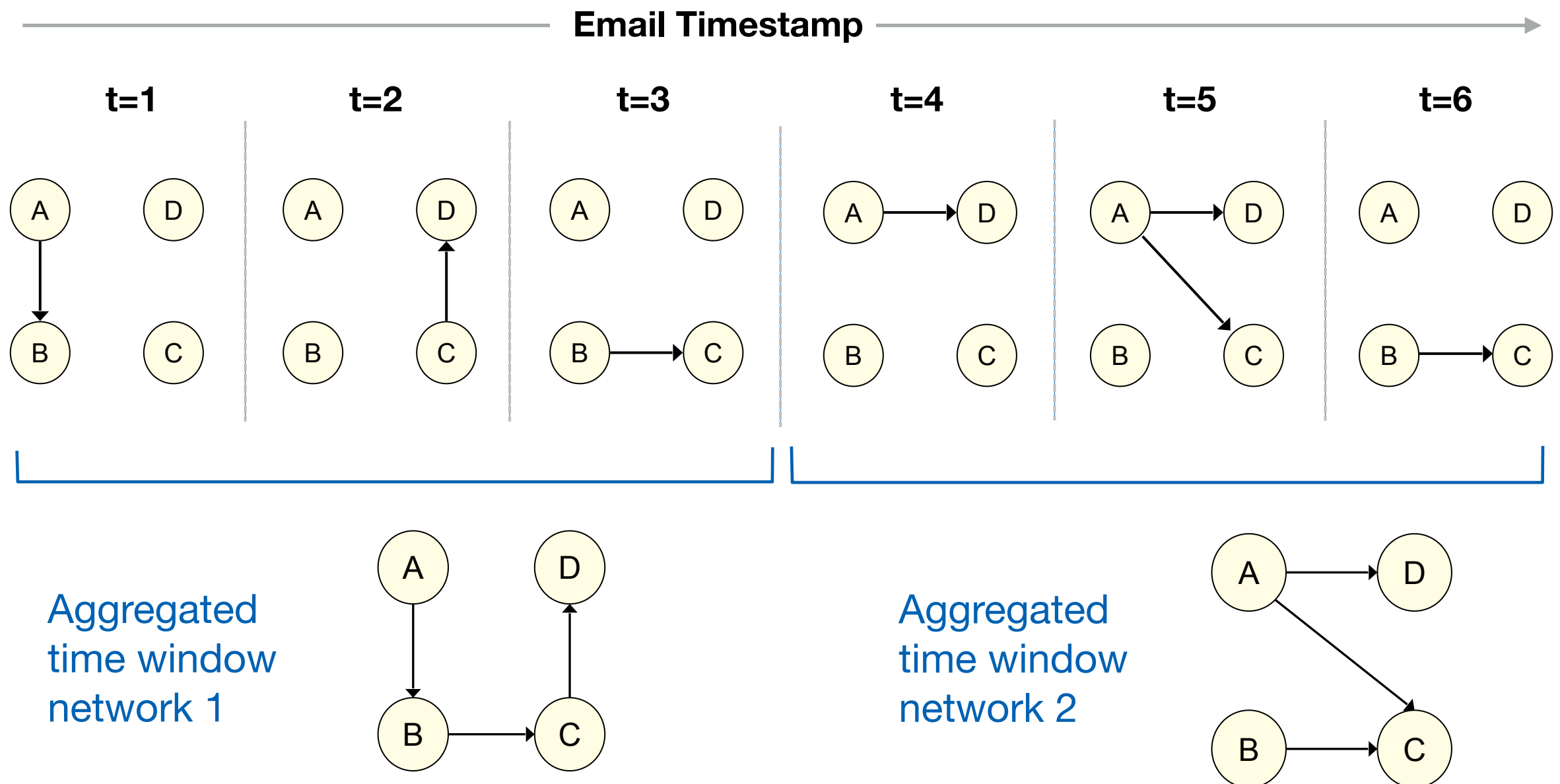


Time Window Network 2



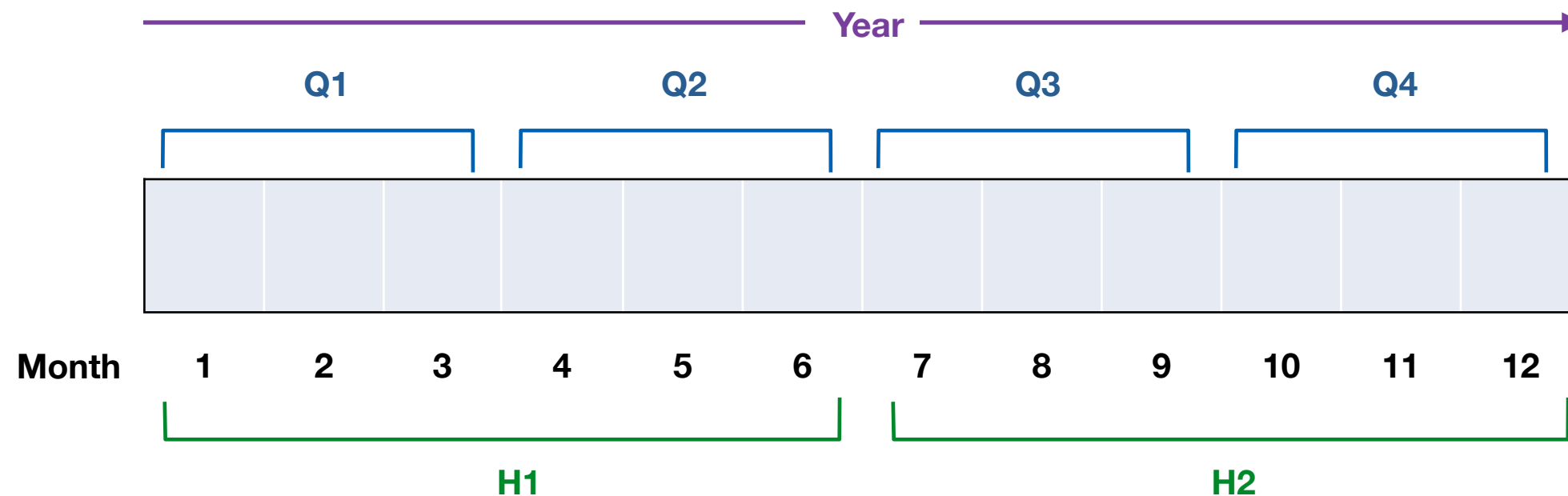
Example: Time Window Networks

- A sequence of emails between four individuals over time can be represented in different ways. We could analyse the emails at six individual time points, or group them into two aggregated time windows, each covering three time steps.



Time Window Size

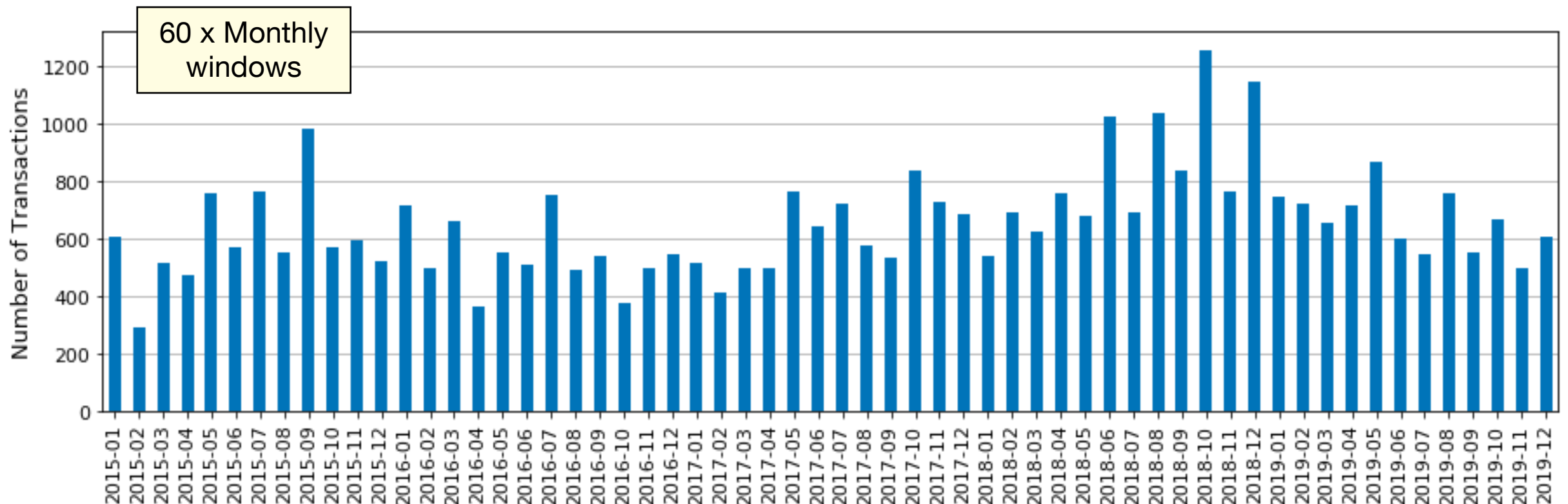
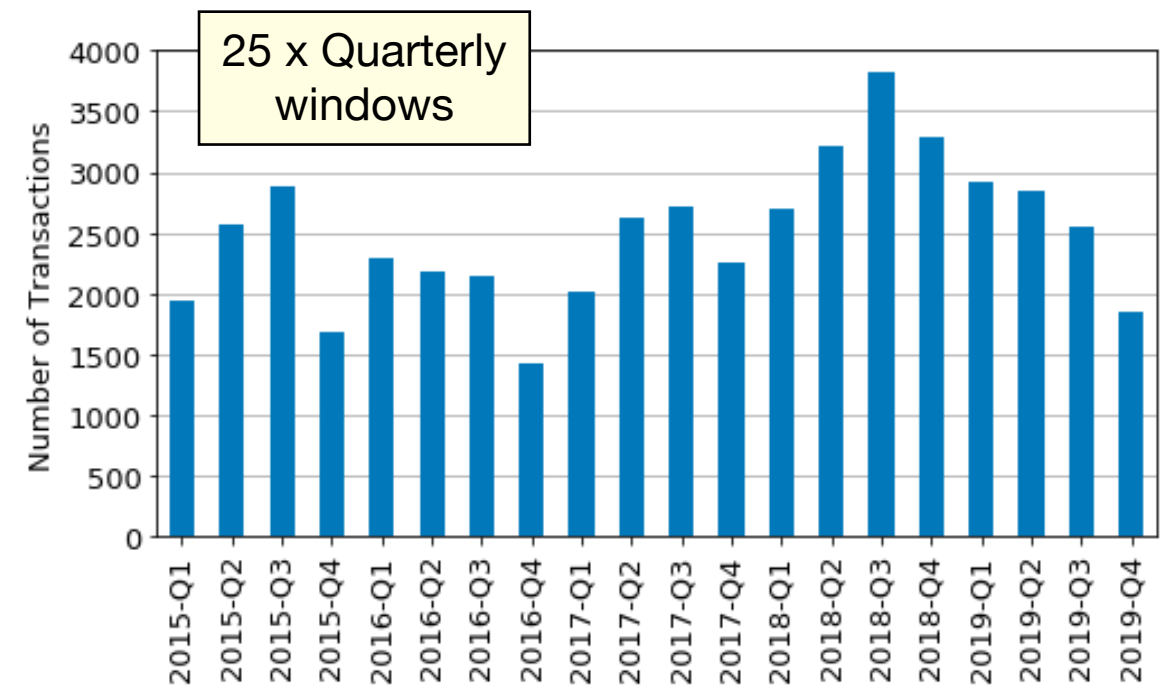
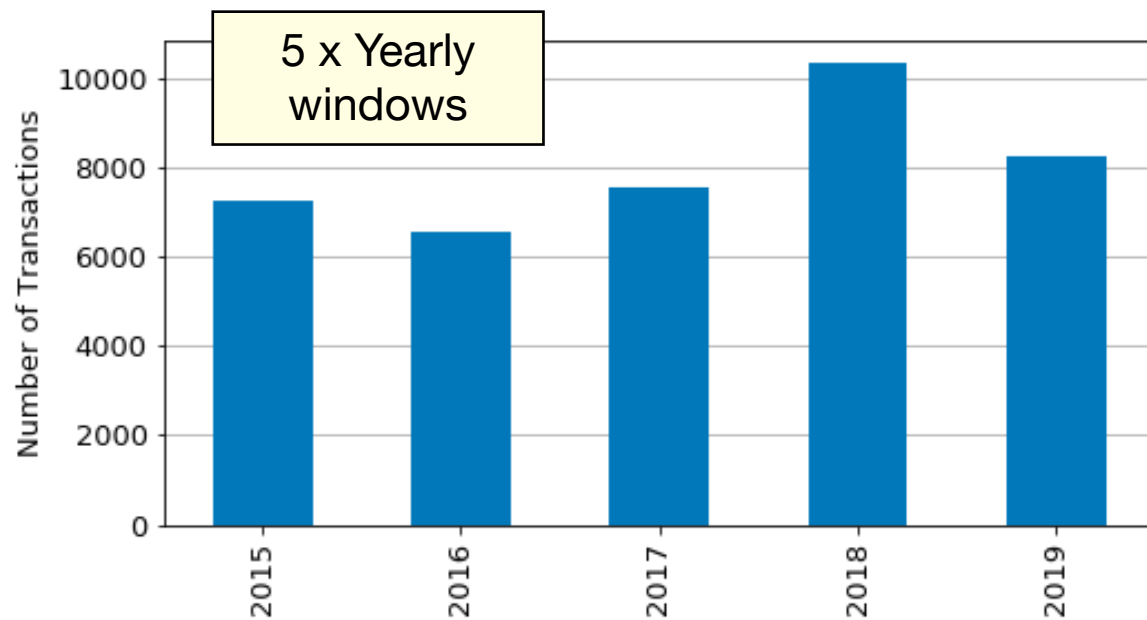
- When creating dynamic networks using time windows, a key decision relates to the **window size** to divide the data.



- This choice determines the **temporal resolution** of the analysis. The appropriate window size depends on the nature of the data and task being performed.
 - If the window is too small, each snapshot may contain too little data and appear noisy.
 - If the window is too large, noise is removed, but the data is also over-smoothed, so useful temporal information is lost.

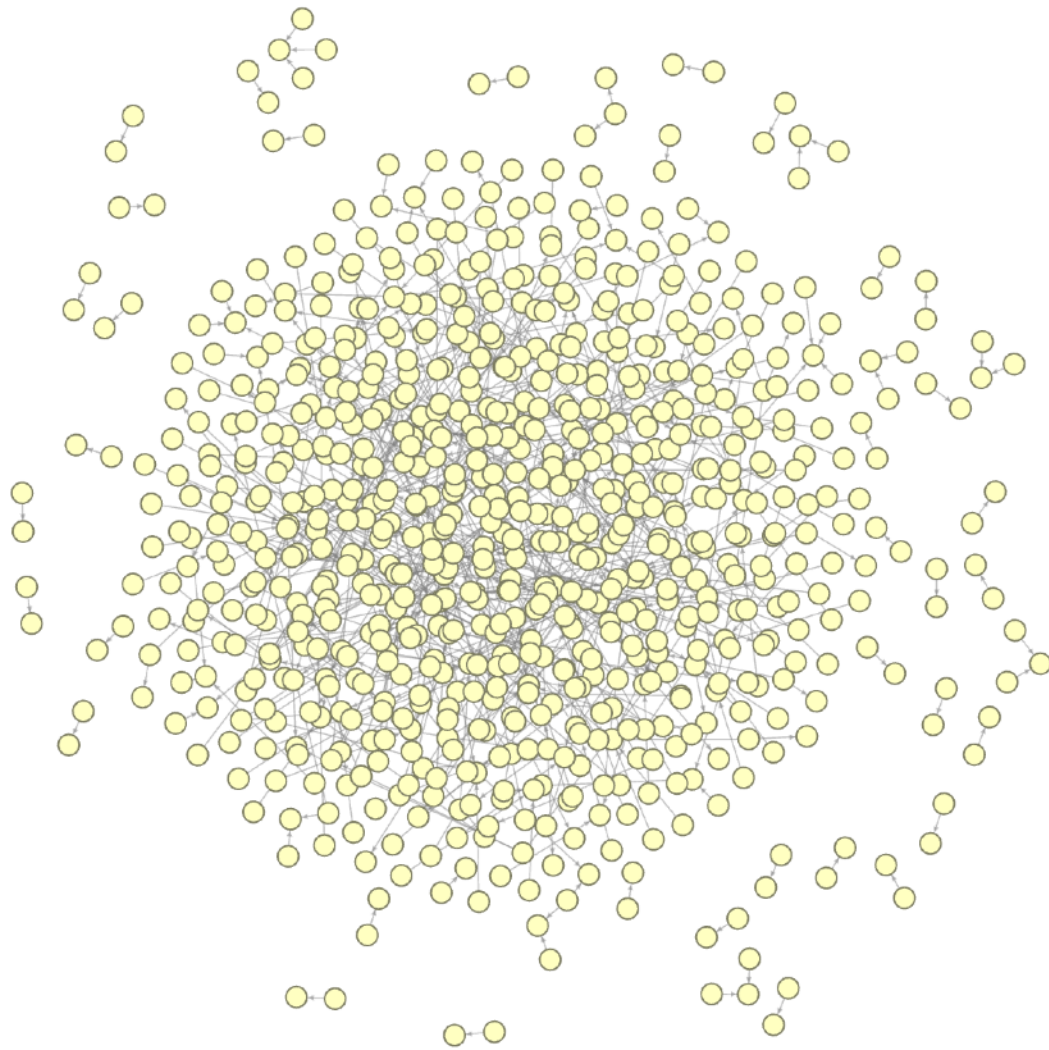
Time Window Size

- Example:** Dataset of 40,000 timestamped financial transactions.

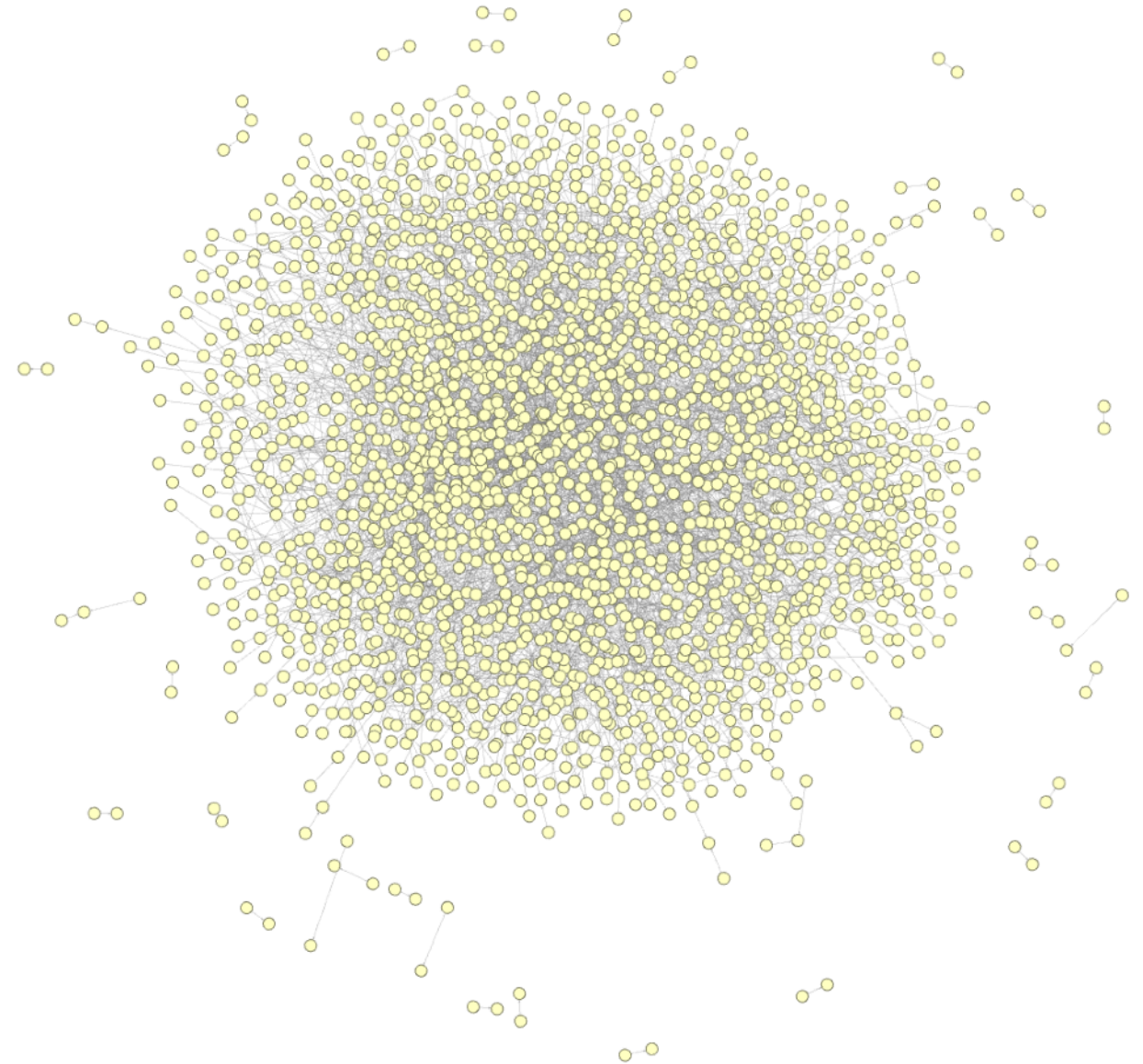


Time Window Size

- **Example:** Dataset of 40,000 timestamped financial transactions.



Monthly Time Windows
Window: Jan-2015
601 nodes, 763 edges

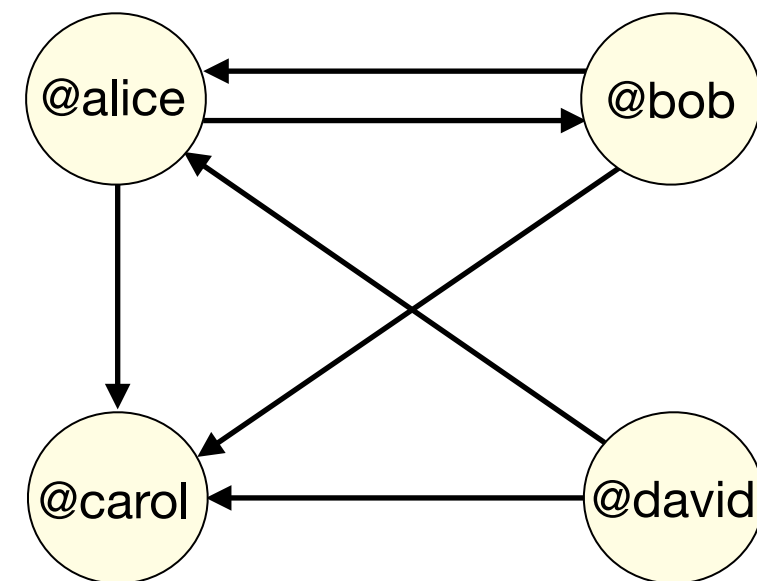


Quarterly Time Windows
Window: 2015-Q1
2240 nodes, 1943 edges

Example: Dynamic Social Media Networks

- In the context of social media platforms, we might want to analyse how user behaviour is changing over time.
- For instance, on Twitter/X we could study how follower relations change over time - i.e., when users follow (or unfollow) other users.

Date	Follower	Followee
13/01/2023	@bob	@alice
13/01/2023	@alice	@bob
28/01/2023	@david	@alice
09/02/2023	@alice	@carol
11/02/2023	@bob	@carol
12/02/2023	@david	@carol

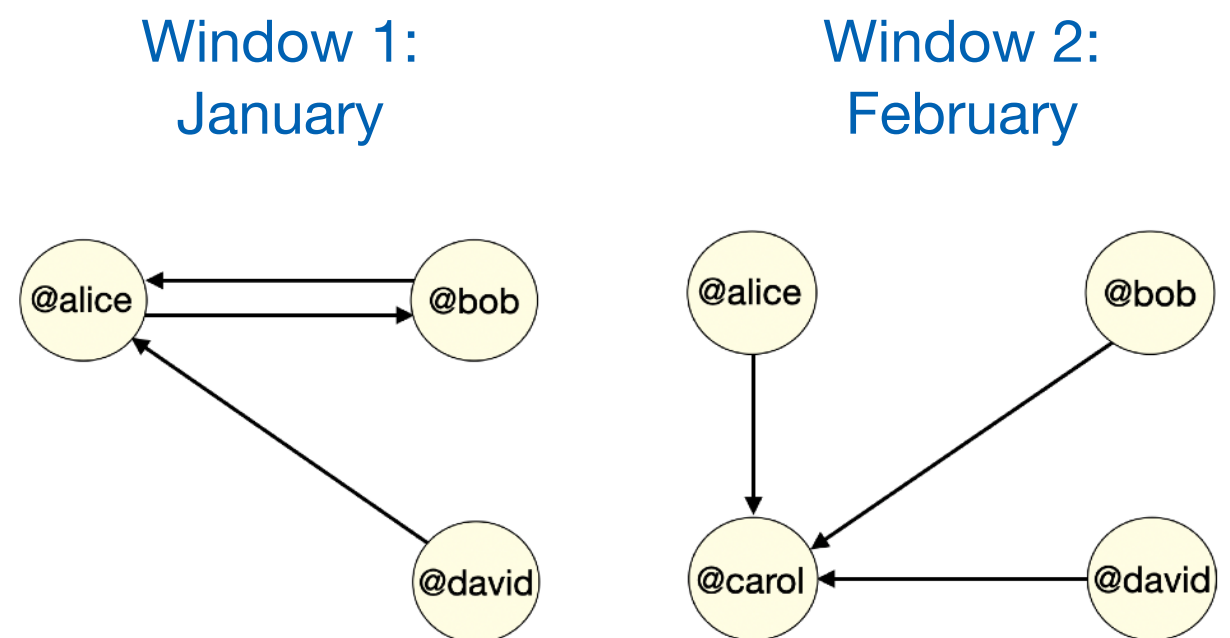


Evolution of follower network
for four users, edge by edge

Example: Dynamic Social Media Networks

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- For instance, on Twitter/X we could study how follower relations change over time - i.e., when users follow (or unfollow) other users.

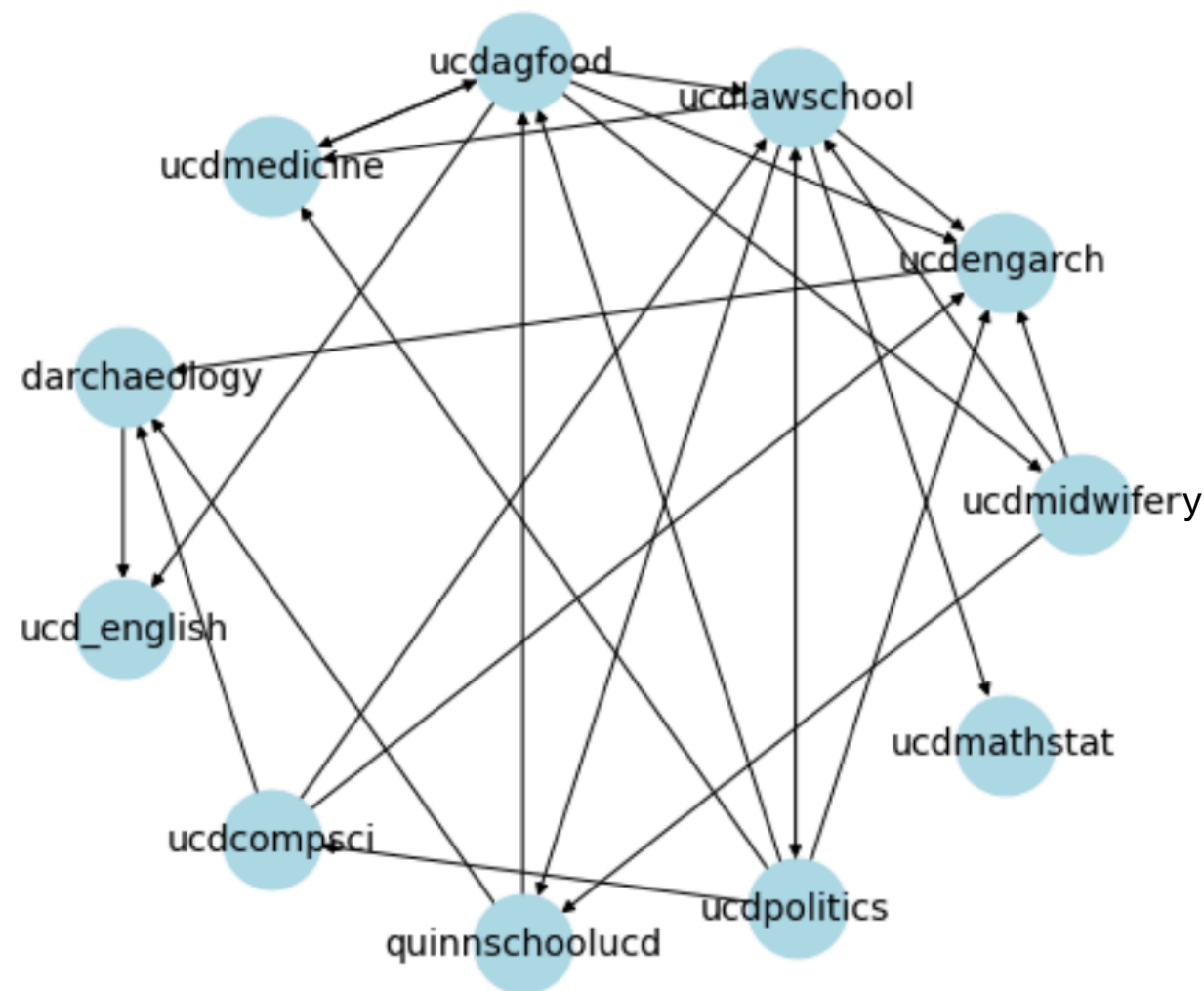
Date	Follower	Followee
13/01/2023	@bob	@alice
13/01/2023	@alice	@bob
28/01/2023	@david	@alice
09/02/2023	@alice	@carol
11/02/2023	@bob	@carol
12/02/2023	@david	@carol



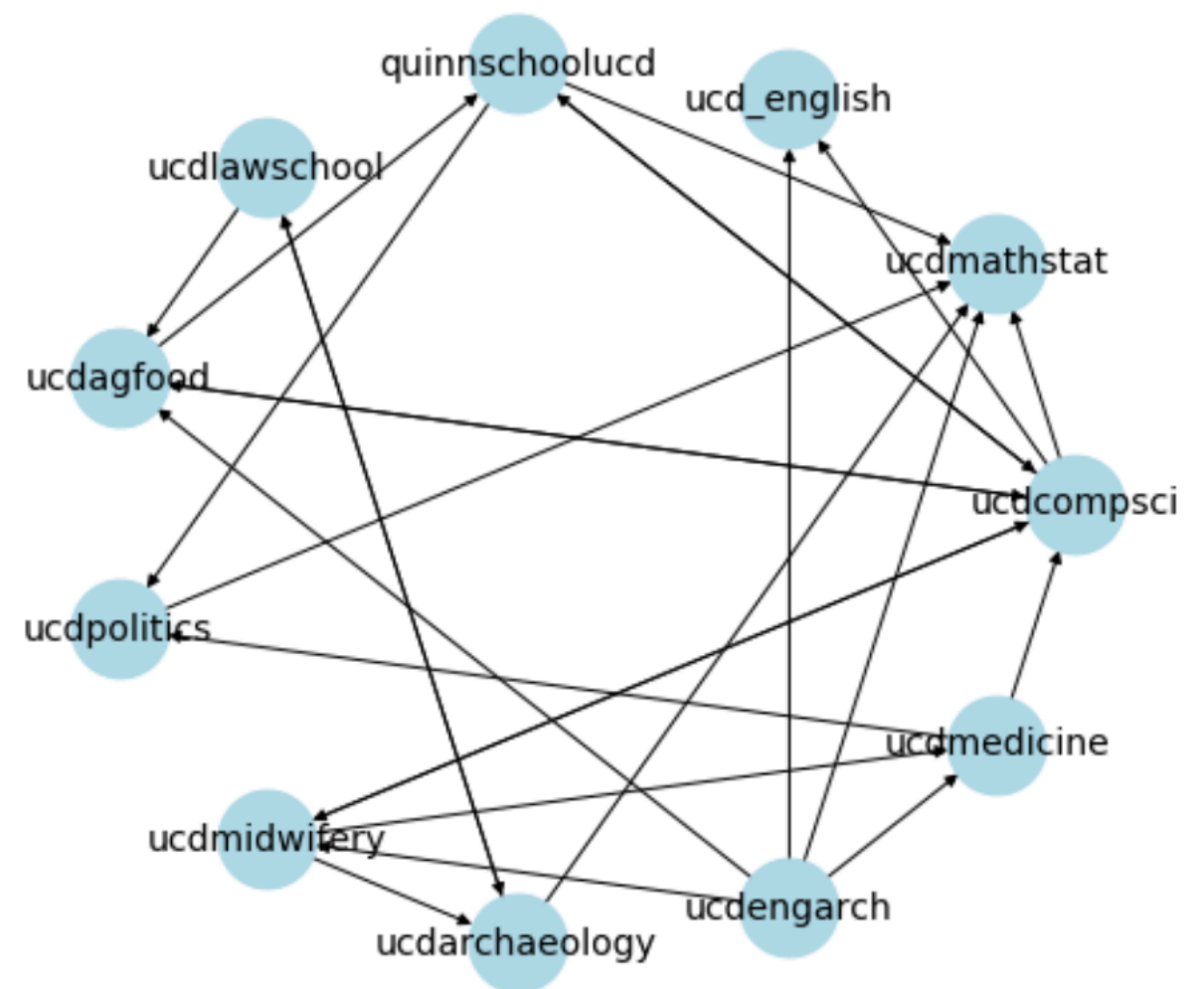
Dynamic follower network, divided
into two monthly time windows

Example: Dynamic Social Media Networks

- Example:** Window networks generated from timestamped follower data for 12 Twitter/X accounts. Directed edges indicate when user *A* started following user *B*.



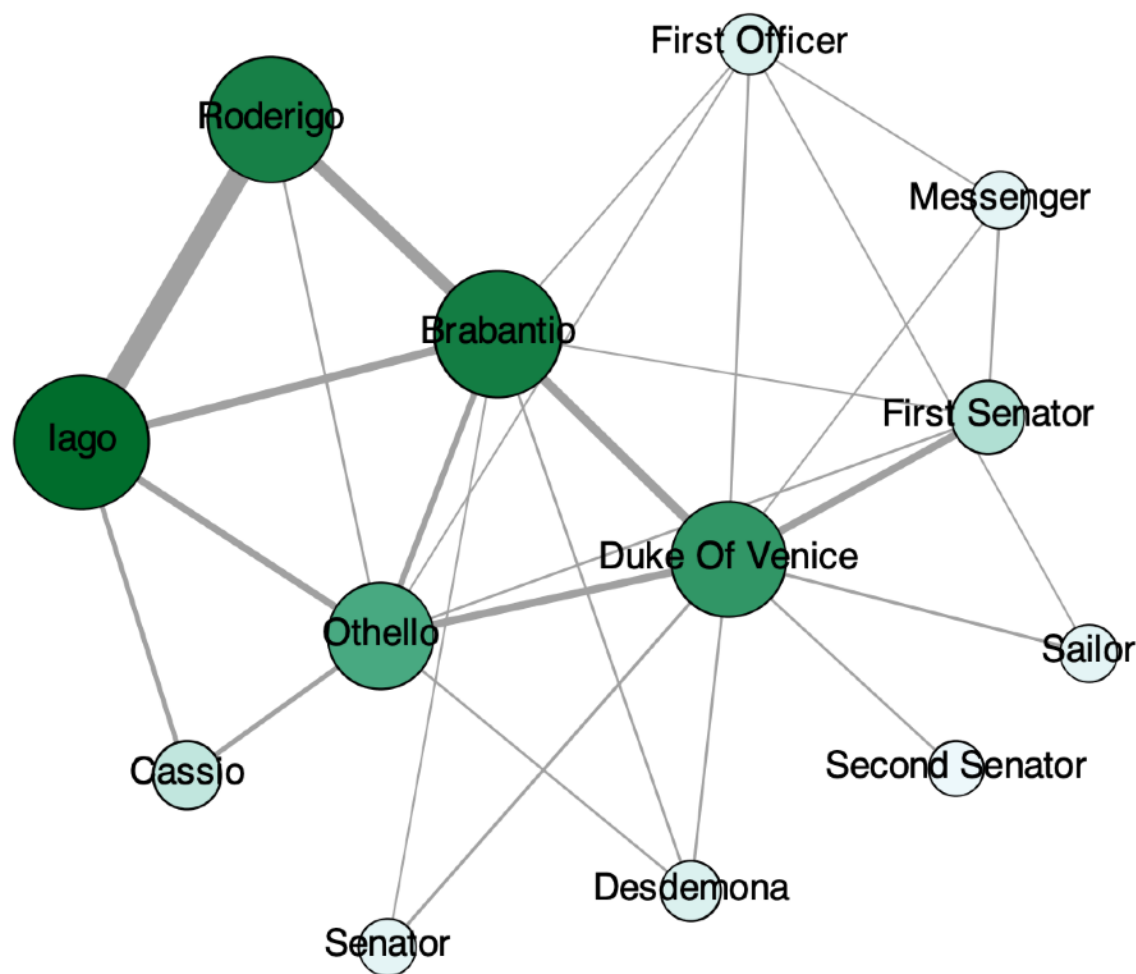
Time window: 2020



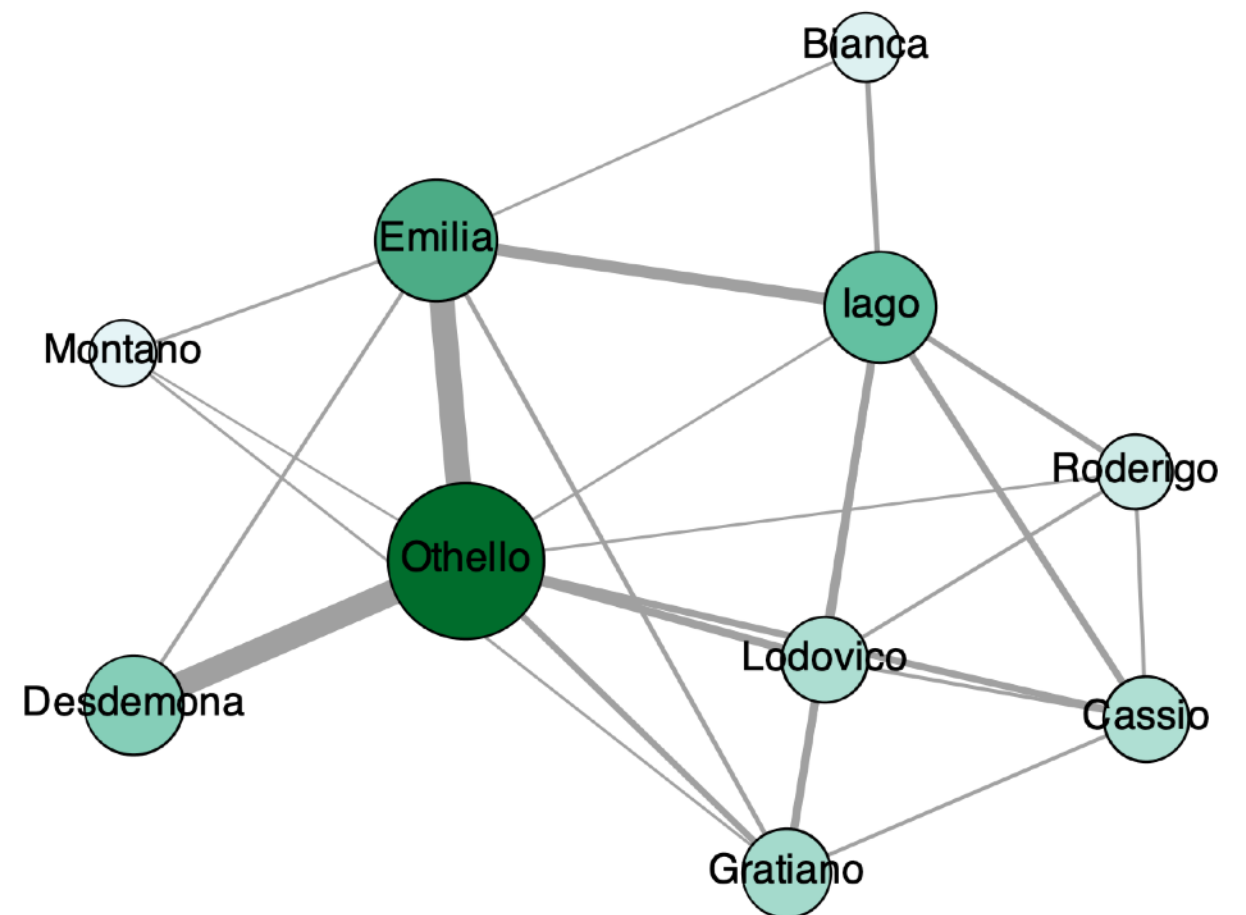
Time window: 2021

Example: Character Networks

- When studying character networks in a drama, each act can be treated as a time window, with a separate network for each act.
- **Example:** Act-based networks for Shakespeare's *Othello*.



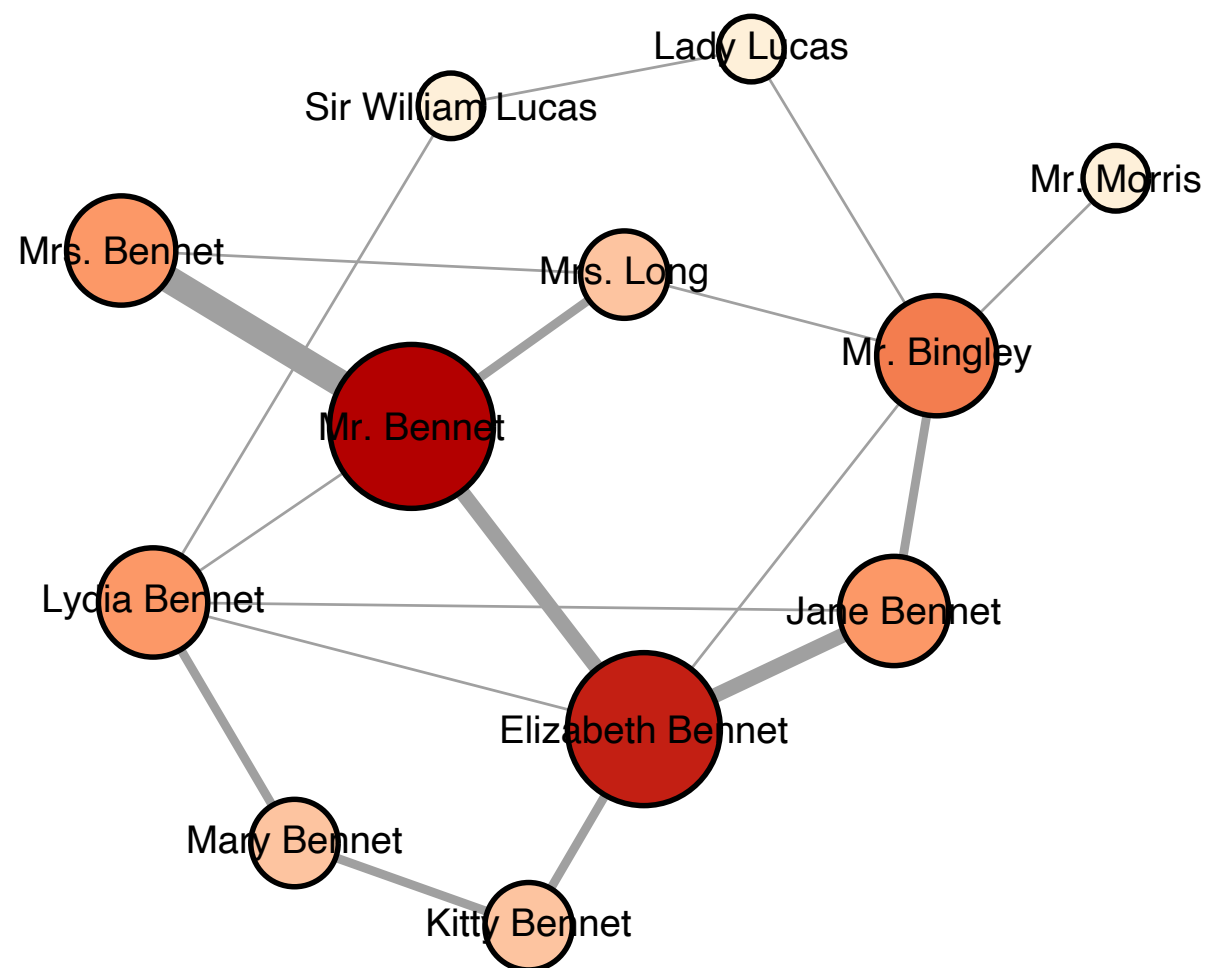
ACT 1



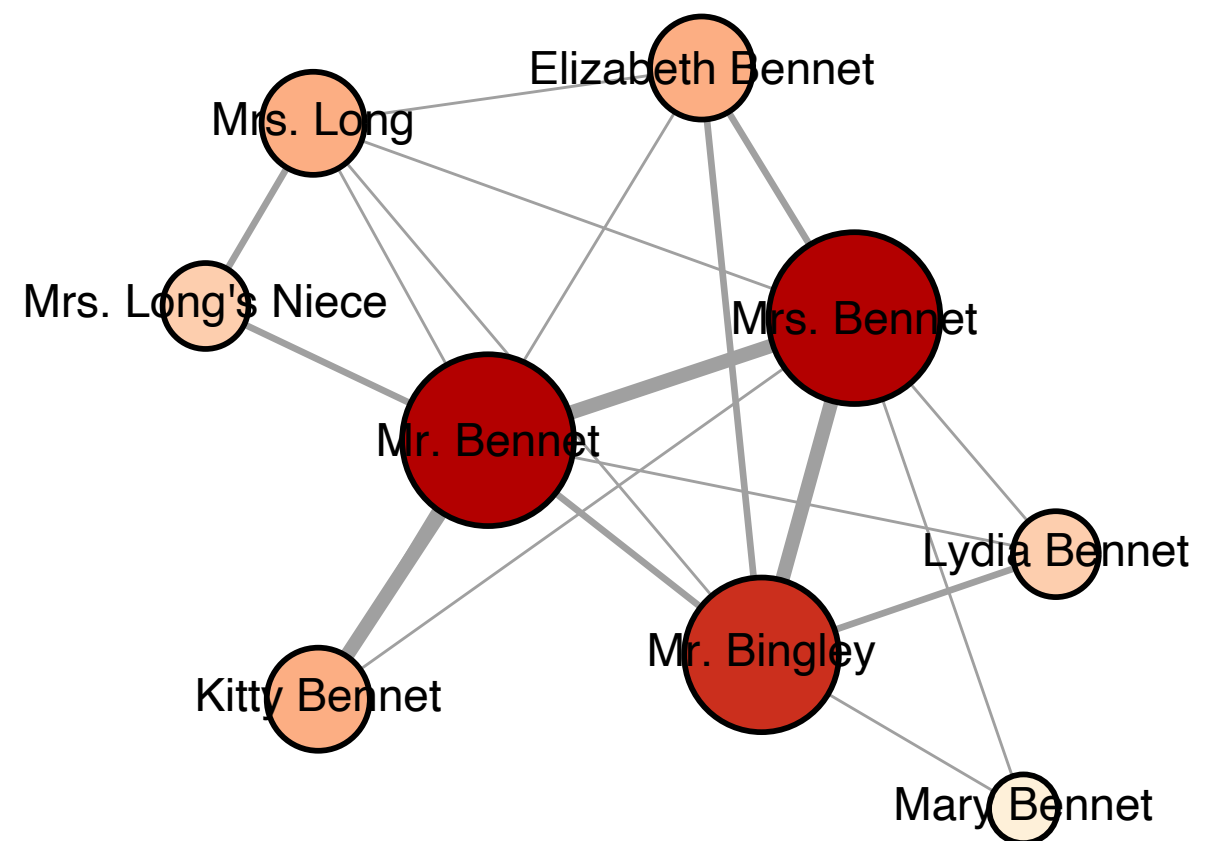
ACT 5

Example: Character Networks

- Similarly, when studying character networks extracted from novels, each chapter can be treated as a separate time window.
- **Example:** Chapter networks for Jane Austen's *Pride and Prejudice*.



Chapter 1



Chapter 2

Creating Dynamic Networks

- Two slightly different methods for creating a dynamic network.
- **Method 1:**
 1. Sort the rows of the dataset by timestamp.
 2. Specify a **time window size** s (e.g. day, month, year). Divide the timestamped data into windows of equal duration s .
 3. Construct window networks, each based on the data which falls within a given time window.
- **Method 2:**
 1. Sort the rows of the dataset by timestamp.
 2. Specify a **fixed number of time windows** n (e.g. $n=10$). Divide the timestamped data into n windows, all of equal length. The actual window size will depend on the time span of the data.
 3. Construct window networks, each based on the data which falls within a given time window.

Creating Dynamic Networks - Method 1

- **Example:** Data for 19 Zoom calls between 9 pairs of individuals.

month	num	caller1	caller2
Jan	1	Oscar	Poppy
Feb	2	Emma	Poppy
Mar	3	Ethan	Emily
Mar	3	Alex	Poppy
Apr	4	Daniel	Emma
Apr	4	Harry	Alex
May	5	Emily	Harry
May	5	Oscar	Sophia
Jun	6	Ethan	Poppy
Jun	6	Emily	Ethan
Jul	7	Emily	Poppy
Jul	7	Emma	Poppy
Aug	8	Alex	Sophia
Aug	8	Emma	Emily
Sep	9	Ethan	Harry
Oct	10	Harry	Poppy
Nov	11	Poppy	Daniel
Nov	11	Harry	Emily
Dec	12	Oscar	Emma

- Specify a time window duration of a quarter.
- The span of the data is 12 months, so this will yield $12/3 = 4$ time windows.
- Split the data into these time windows...

Window 1 (Q1)

Jan	1	Oscar	Poppy
Feb	2	Emma	Poppy
Mar	3	Ethan	Emily
Mar	3	Alex	Poppy

Window 2 (Q2)

Apr	4	Daniel	Emma
Apr	4	Harry	Alex
May	5	Emily	Harry
May	5	Oscar	Sophia
Jun	6	Ethan	Poppy
Jun	6	Emily	Ethan

Window 3 (Q3)

Jul	7	Emily	Poppy
Jul	7	Emma	Poppy
Aug	8	Alex	Sophia
Aug	8	Emma	Emily
Sep	9	Ethan	Harry

Window 4 (Q4)

Oct	10	Harry	Poppy
Nov	11	Poppy	Daniel
Nov	11	Harry	Emily
Dec	12	Oscar	Emma

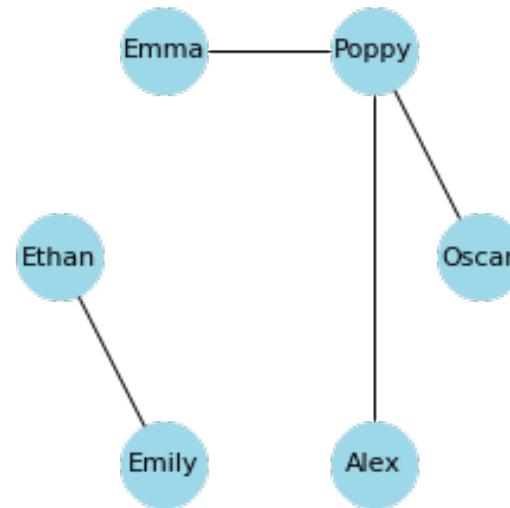
Creating Dynamic Networks - Method 1

- **Example:** Data for 19 Zoom calls between 9 pairs of individuals.

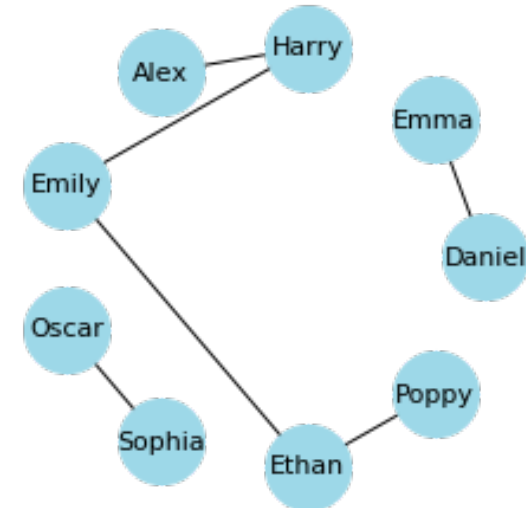
month	num	caller1	caller2
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Feb	2	Emma	Poppy
Mar	3	Ethan	Emily
Mar	3	Alex	Poppy
Apr	4	Daniel	Emma
Apr	4	Harry	Alex
May	5	Emily	Harry
May	5	Oscar	Sophia
Jun	6	Ethan	Poppy
Jun	6	Emily	Ethan
Jul	7	Emily	Poppy
Jul	7	Emma	Poppy
Aug	8	Alex	Sophia
Aug	8	Emma	Emily
Sep	9	Ethan	Harry
Oct	10	Harry	Poppy
Nov	11	Poppy	Daniel
Nov	11	Harry	Emily
Dec	12	Oscar	Emma

- Based on the time window data subsets, we can create a sequence of four window networks.

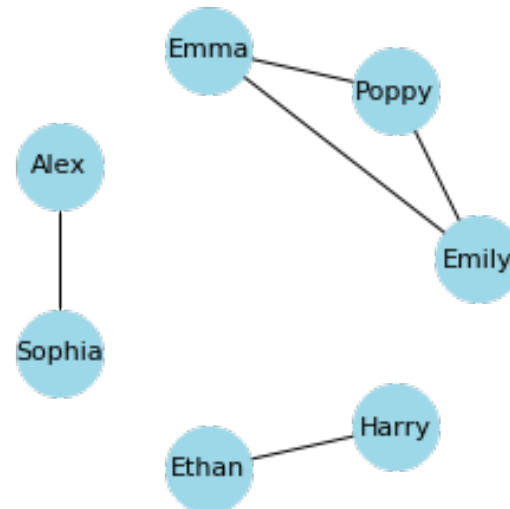
Window 1
(Q1)



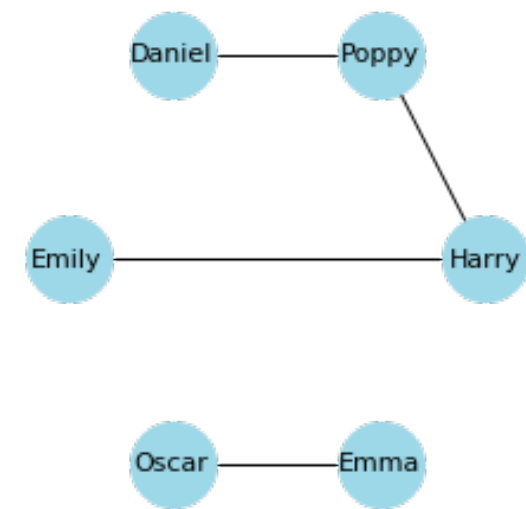
Window 2
(Q2)



Window 3
(Q3)



Window 4
(Q4)



Creating Dynamic Networks - Method 2

- **Example:** Data for 19 Zoom calls between 9 pairs of individuals.

month	num	caller1	caller2
Jan	1	Oscar	Poppy
Feb	2	Emma	Poppy
Mar	3	Ethan	Emily
Mar	3	Alex	Poppy
Apr	4	Daniel	Emma
Apr	4	Harry	Alex
May	5	Emily	Harry
May	5	Oscar	Sophia
Jun	6	Ethan	Poppy
Jun	6	Emily	Ethan
Jul	7	Emily	Poppy
Jul	7	Emma	Poppy
Aug	8	Alex	Sophia
Aug	8	Emma	Emily
Sep	9	Ethan	Harry
Oct	10	Harry	Poppy
Nov	11	Poppy	Daniel
Nov	11	Harry	Emily
Dec	12	Oscar	Emma

- Specify a number of time windows, say $n=3$.
- The span of the data is 12 months, so this will naturally yield windows of duration 4 months.
- Split the data into three time windows.

Time Window 1

Jan	1	Oscar	Poppy
Feb	2	Emma	Poppy
Mar	3	Ethan	Emily
Mar	3	Alex	Poppy
Apr	4	Daniel	Emma
Apr	4	Harry	Alex

Time Window 2

May	5	Emily	Harry
May	5	Oscar	Sophia
Jun	6	Ethan	Poppy
Jun	6	Emily	Ethan
Jul	7	Emily	Poppy
Jul	7	Emma	Poppy
Aug	8	Alex	Sophia
Aug	8	Emma	Emily

Time Window 3

Sep	9	Ethan	Harry
Oct	10	Harry	Poppy
Nov	11	Poppy	Daniel
Nov	11	Harry	Emily
Dec	12	Oscar	Emma

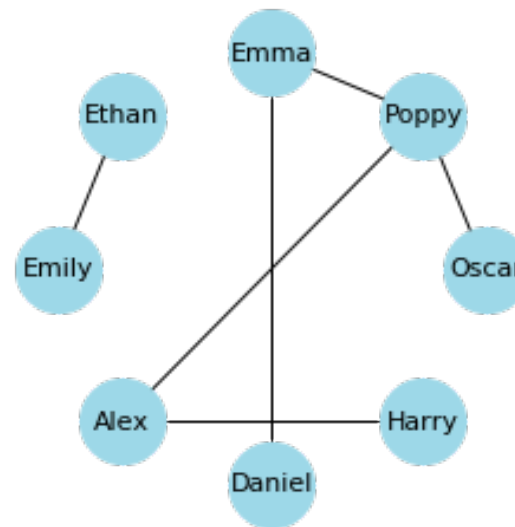
Creating Dynamic Networks - Method 2

- **Example:** Data for 19 Zoom calls between 9 pairs of individuals.

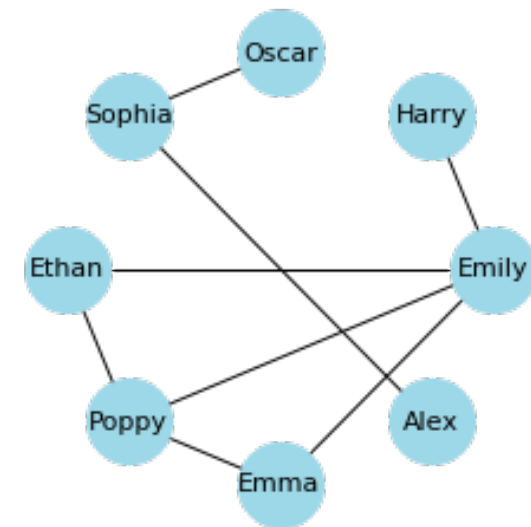
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Jan	1	Oscar	Poppy
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Jun	6	Ethan	Poppy
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- Based on the time window data subsets, we can create a sequence of three window networks.

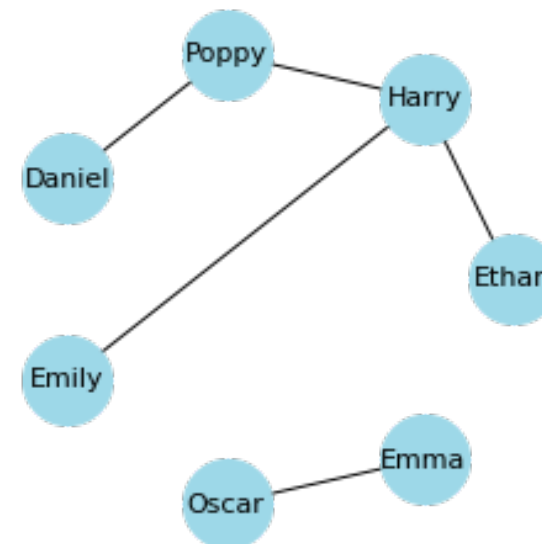
Window
Network 1



Window
Network 2



Window
Network 3



Creating Dynamic Networks in Python

- We can implement the time window generation and network construction steps using Pandas and NetworkX.

Read the full timestamped dataset into a DataFrame

```
df = pd.read_csv("dynamic-presence.csv", sep="\t")
```

Apply Method 1 to divide the data into four quarterly time windows

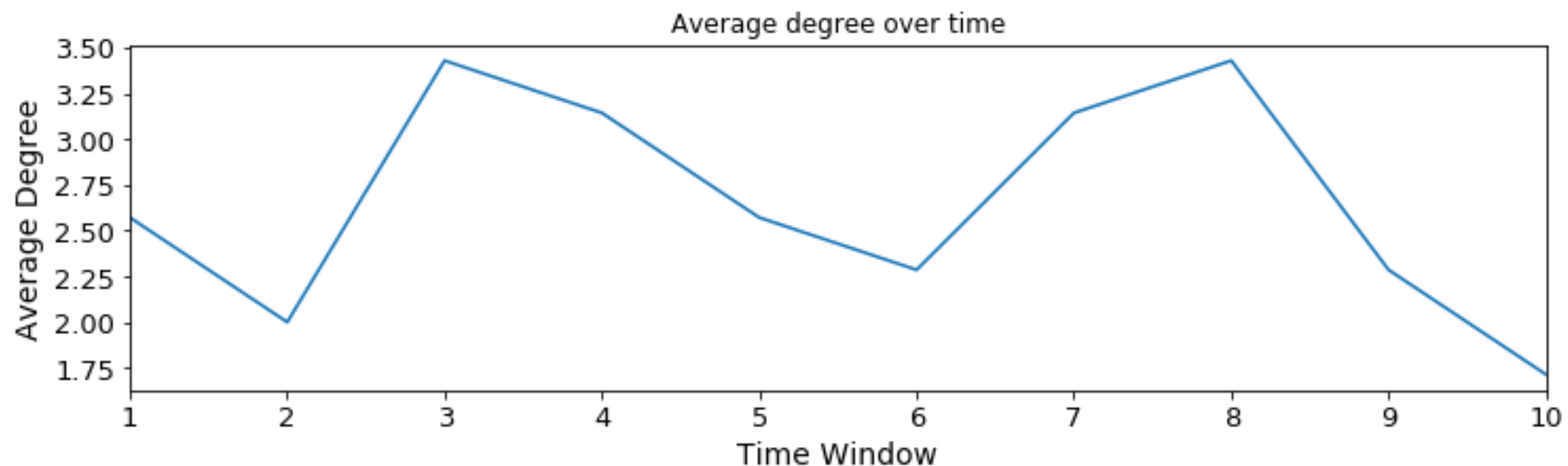
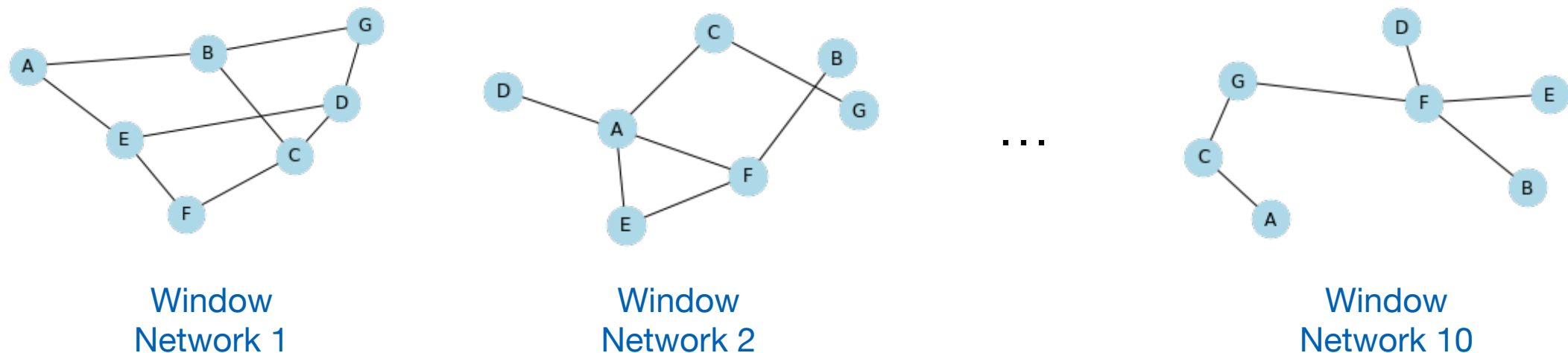
```
window_frames = []
month_start = 1
for quarter in range(1,5):
    wf = df[(df["num"]>=month_start) & (df["num"]<=month_start+2)]
    window_frames.append( wf )
    month_start += 3
```

Construct four quarterly time window networks

```
window_networks = []
for wf in window_frames:
    g = nx.from_pandas_edgelist(wf, "caller1", "caller2")
    window_networks.append( g )
```

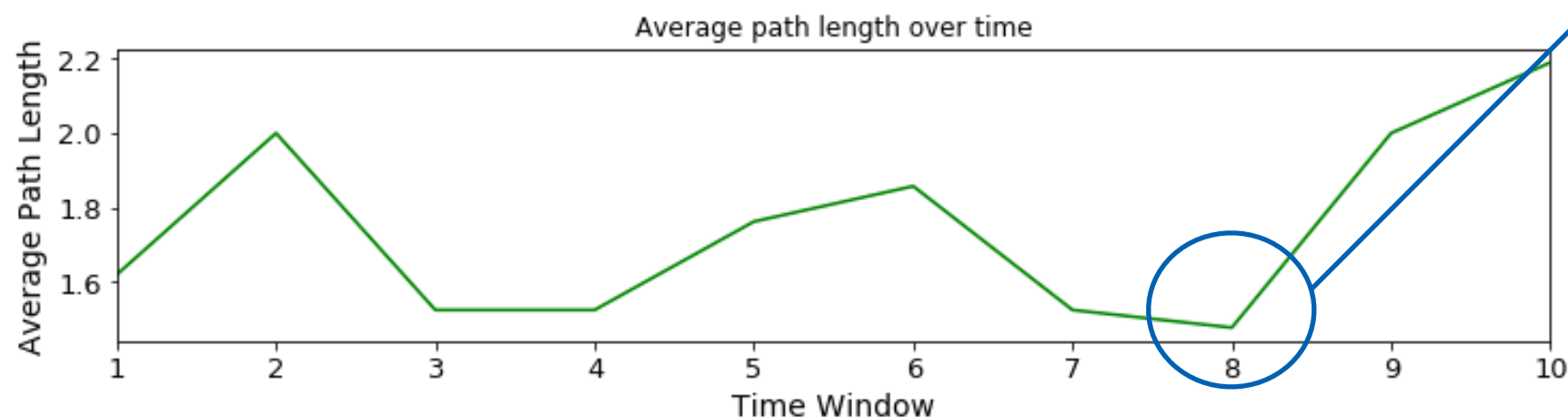
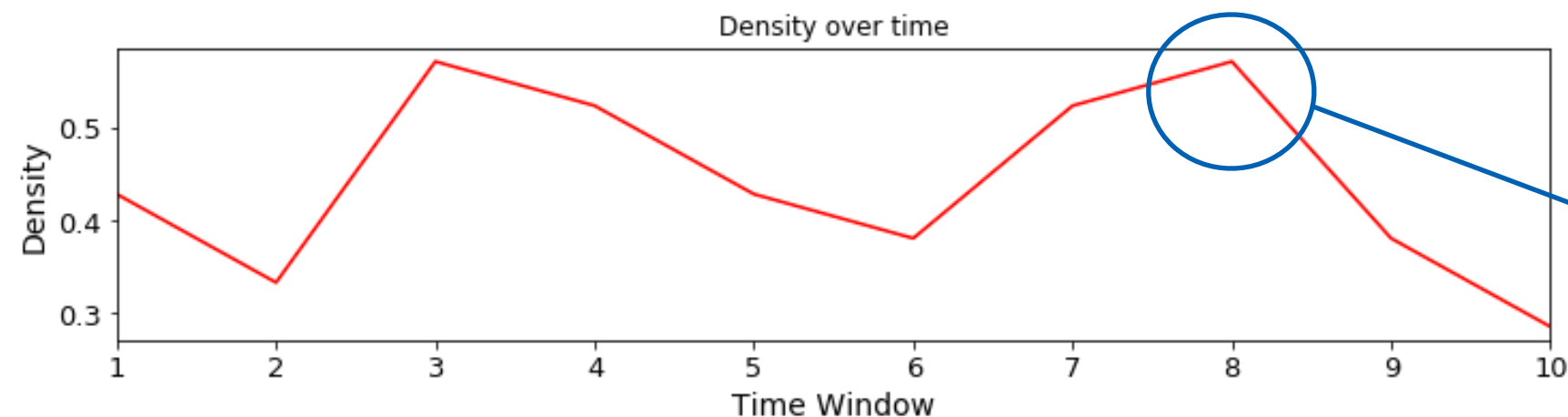
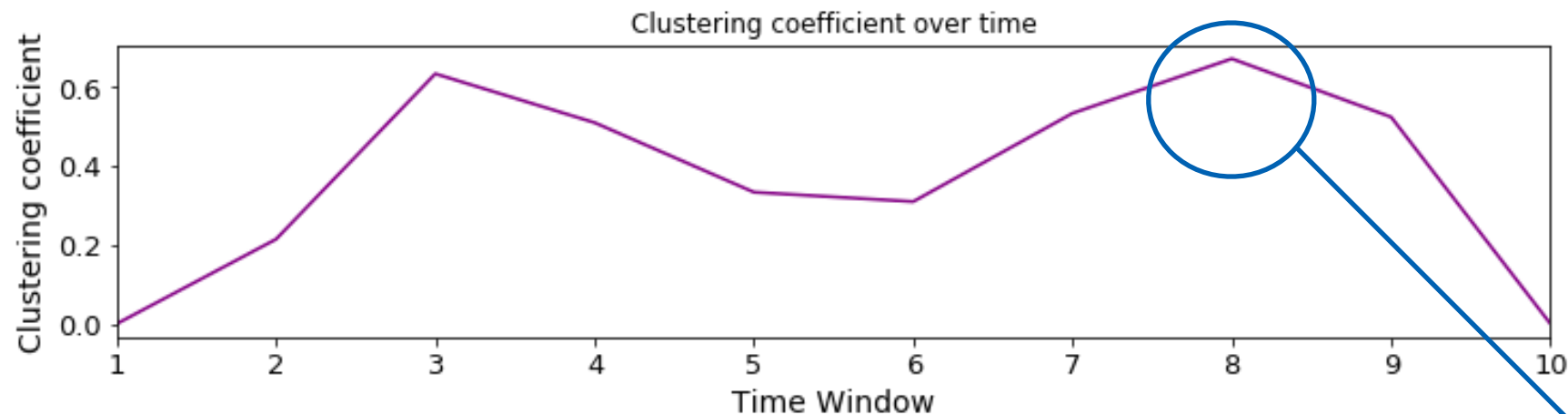
Networks as Time Series

- Once we have constructed a sequence of window networks, we can apply standard network measures to characterise each one.
- This generates a **time series** - i.e., a sequence of values taken at different time points. These points correspond to time windows.



Networks as Time Series

- By applying different measures, we can produce a variety of time series that describe how a network evolves over time.



Potential change point in the dynamic network

Networks as Time Series - Python

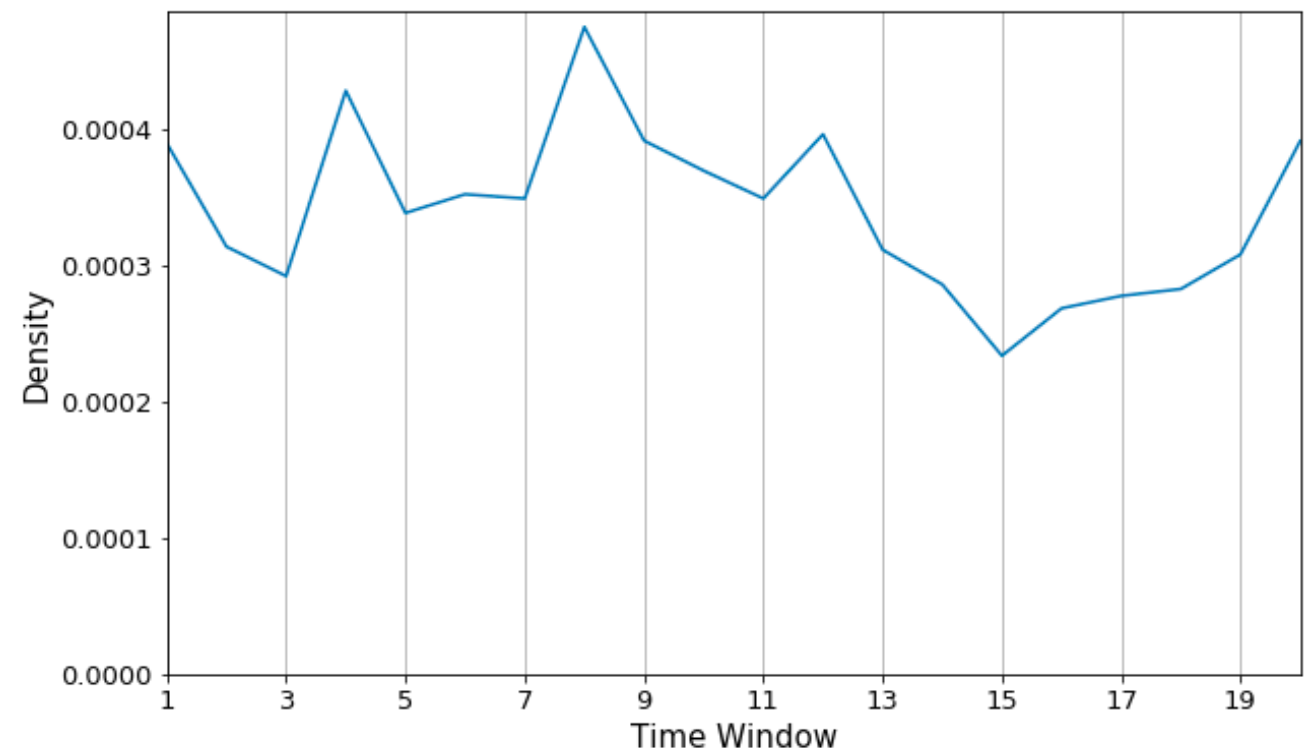
- We can apply NetworkX measures to compute statistics for each time window network and then use Pandas to analyse how these values change over time.

Calculate the density for each time window network and store the results in a Pandas Series

```
values = {}  
for i, g in enumerate(window_networks):  
    values[i+1] = nx.density(g)  
s_density = pd.Series(values, name="Density")
```

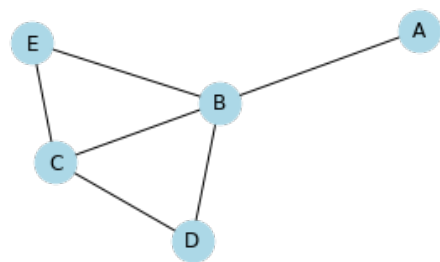
Generate a plot from the Pandas Series

```
ax = s_density.plot(figsize=(10,6))  
ax.set_xlabel("Time Window")  
ax.set_ylabel("Density")  
ax.set_ylim(0)  
ticks = range(1,len(window_ids)+1,2)  
ax.set_xticks(ticks)  
ax.xaxis.grid()  
plt.show()
```

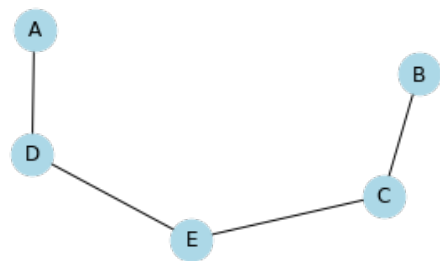


Temporal Node Features

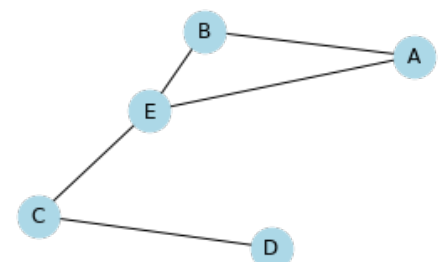
- We can also generate statistics for each node in the time window networks. These can be viewed as **node features**, which describe the behaviour of the nodes over time in a dynamic network.
- These features can be treated either as multiple time series, or stacked as columns in a table.



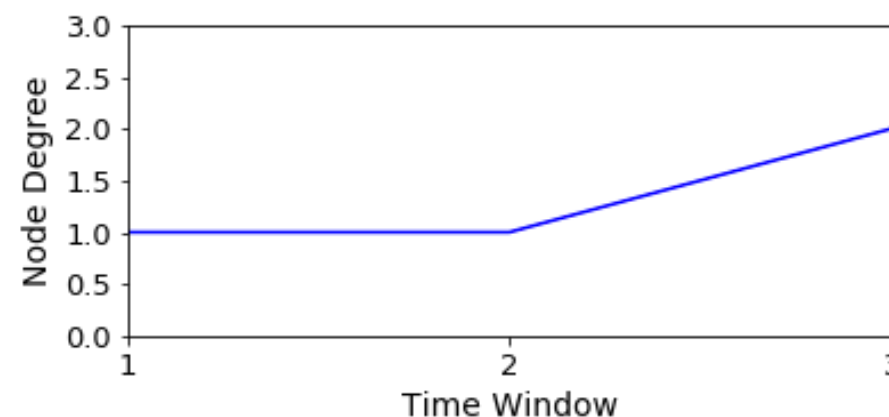
Window Network 1



Window Network 2



Window Network 3



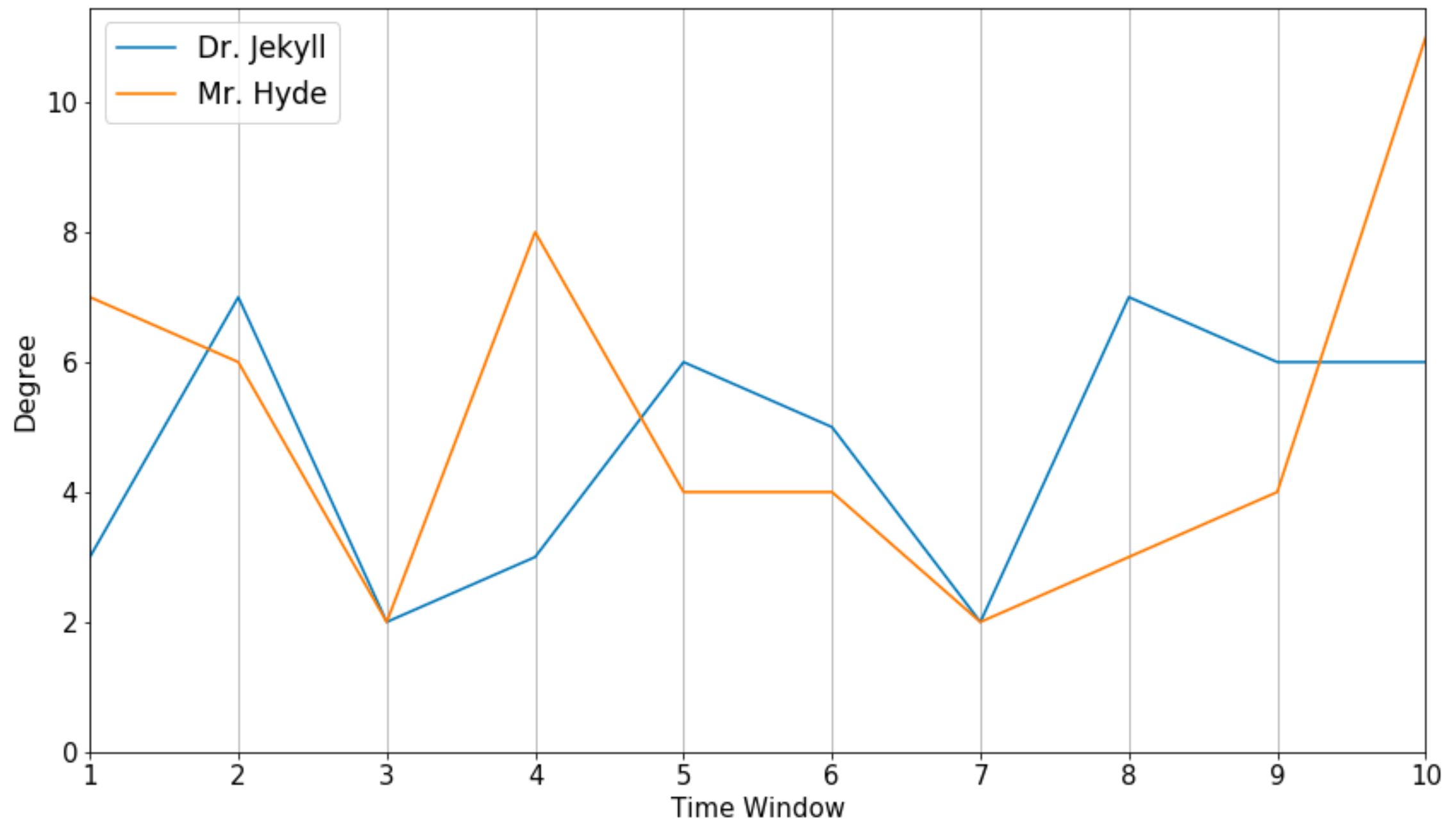
Degree for node A across three time windows

Node	degree_t1	degree_t2	degree_t3
A	1	1	2
B	4	1	2
C	3	2	2
D	2	2	1
E	2	2	3

Table indicating the degree for all nodes, in each of the three windows

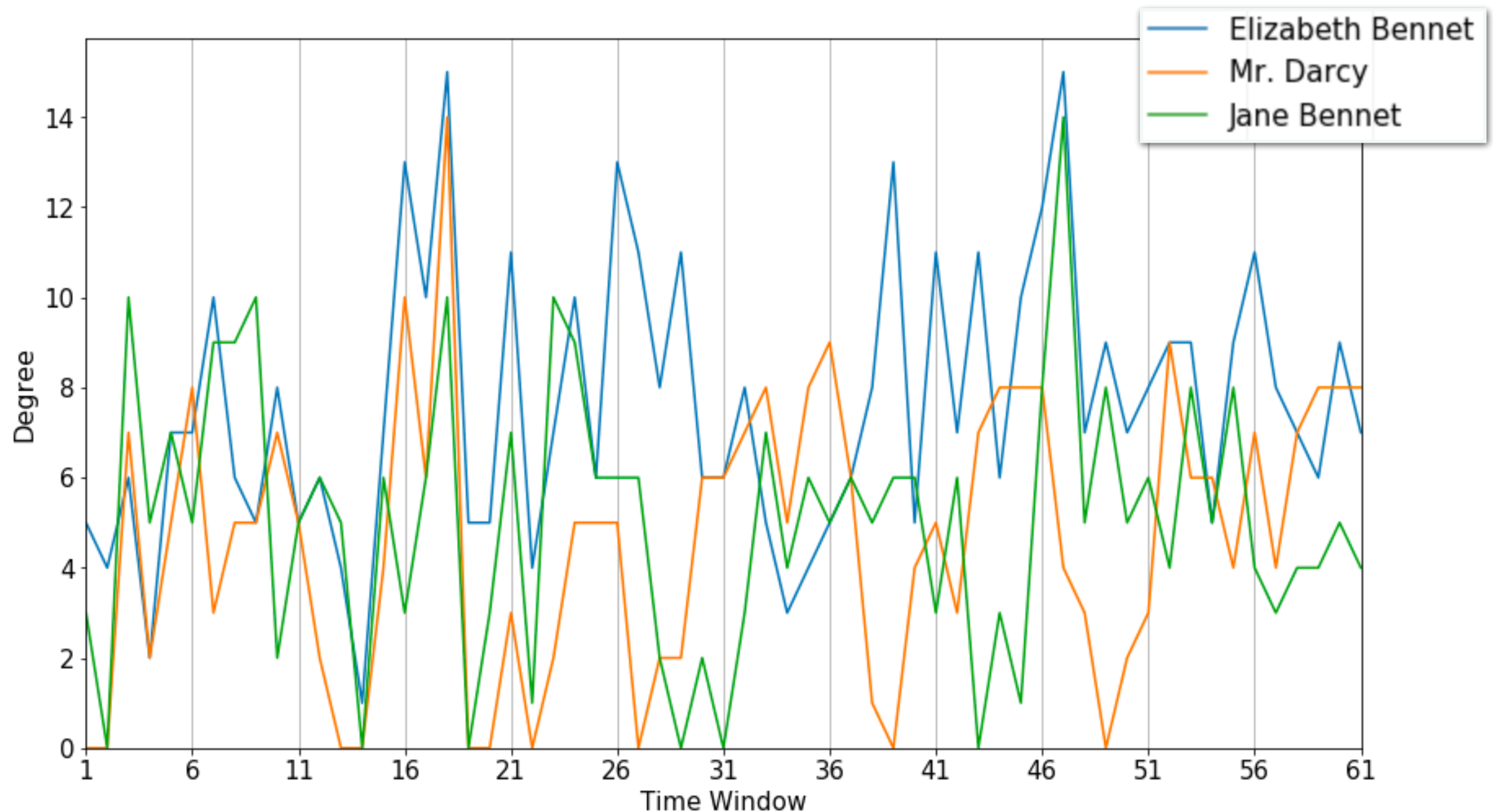
Node Features in Character Networks

- **Example:** Plot of degree for two main characters in 10 chapter networks for Stevenson's *Strange Case of Dr Jekyll and Mr Hyde*.

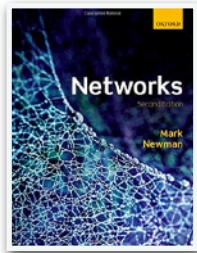


Node Features in Character Networks

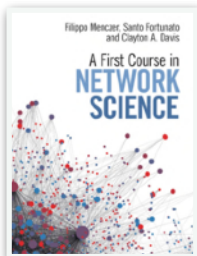
- Example:** Plot of degree for three major characters in 61 chapter networks for Jane Austen's *Pride and Prejudice*.



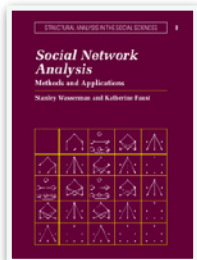
Further Reading



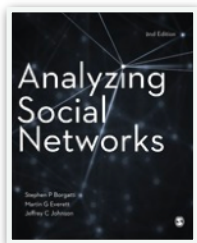
Networks, 2nd Edition
Mark Newman



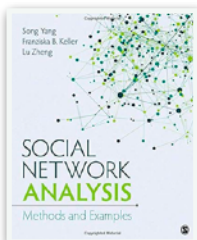
A First Course in Network Science
Filippo Menczer, Santo Fortunato, Clayton A. Davis



Social Network Analysis
Stanley Wasserman, Katherine Faust



Analyzing Social Networks, 2nd Edition
Stephen Borgatti, Martin Everett, Jeffrey Johnson



Social Network Analysis: Methods and Examples
Song Yang Franziska B. Keller, Lu Zheng